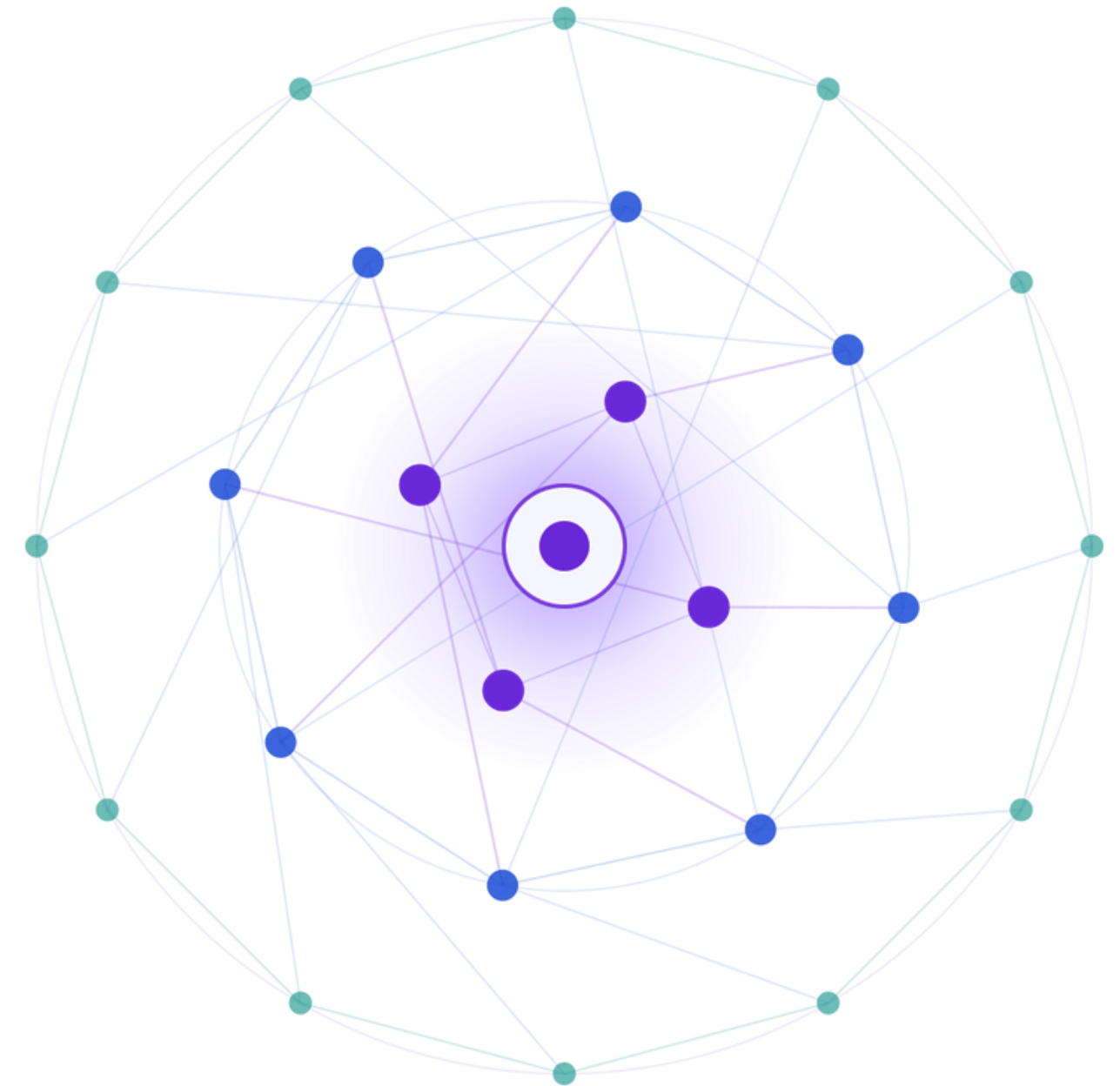


SAMSUNG PRISM

Generative AI Hackathon

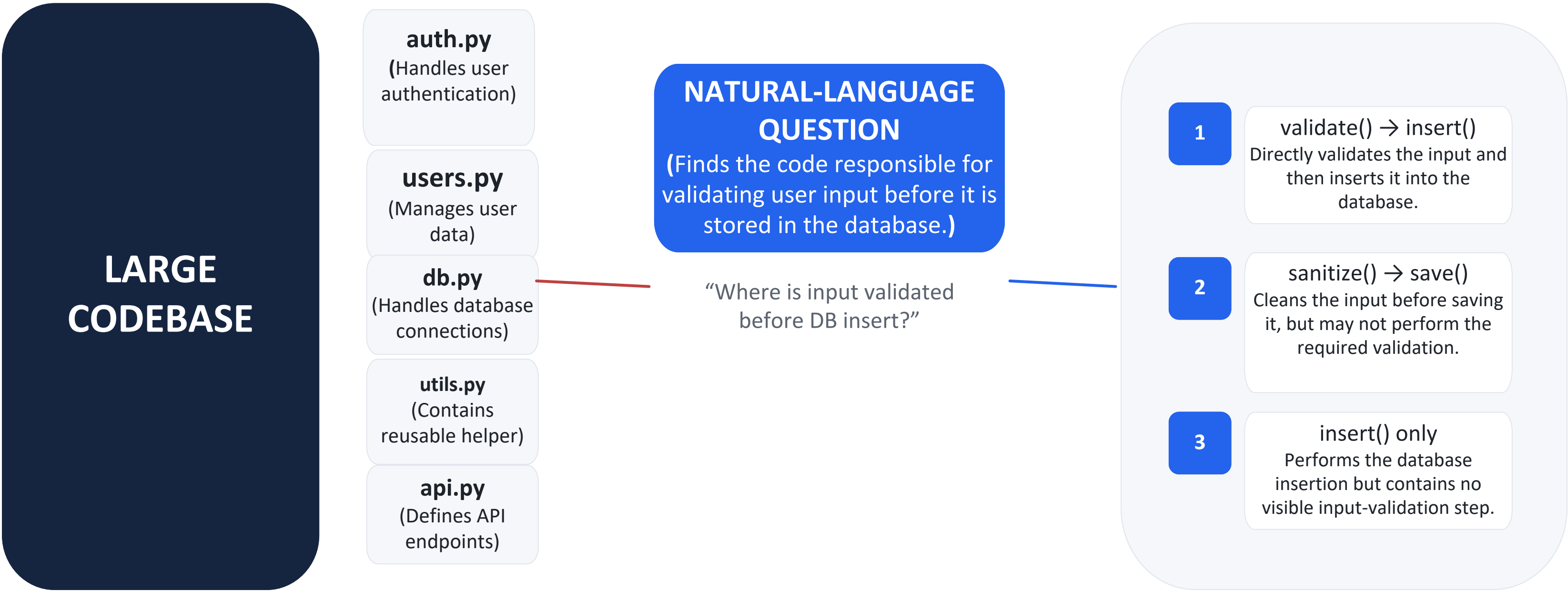
3rd Edition 2026 – 27

- Theme ID – 01: Agentic Code Intelligence
- Team Name – Team Z
- College Name – M. S. Ramaiah Institute of Technology (MSRIT)
- Member 1 – Reddy Janavi P (1ms25ai057@msrit.edu)
- Member 2 – Aditi S Amin (1ms5ai006@msrit.edu)
- Member 3 – Pratheeksha ND (1ms5ai052@msrit.edu)
- Submission GitHub link – [PASTE PUBLIC GITHUB LINK]



Theme & Problem

Why code retrieval becomes the bottleneck for AI coding agents

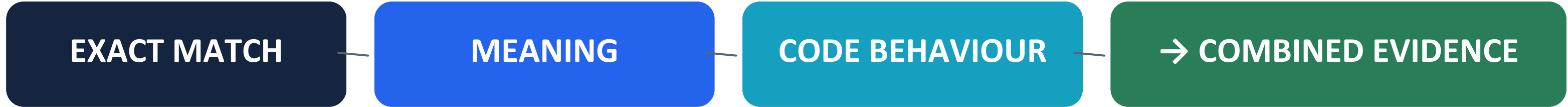


Goal: retrieve the right snippets quickly — before any downstream AI reasoning or generation.

Existing Solutions & Gaps

Why a multi-signal retrieval pipeline is needed

Approach	Strength	Limitation
Keyword / grep	Fast; exact identifiers	Misses semantic matches
Dense embeddings	Understands meaning	Can miss exact APIs / symbols
LLM over repository	Strong reasoning	Slow, costly, context-heavy
Single-pass retrieval	Simple pipeline	Weak ranking / redundancy control
Our hybrid approach	Lexical + semantic + structure + intent	Needs indexing + tuning



Our Solution & Architecture

Multi-stage retrieval: broad recall first, focused ranking second



RETRIEVAL LAYER



Demo & Product Walkthrough

Illustrative example of how the system reasons about a code query

QUERY → “Where is user input validated before database insert?”

normalize(input)
validate(input)
db.insert(input)

Expected relevance: HIGH
Validation occurs before persistence

trim(input)
if locale: check(input)
forward(input)

Expected relevance: MEDIUM
Some preprocessing, weaker DB evidence

db.insert(record)

Expected relevance: LOW
No visible validation



Illustrative ranking — actual scores are produced by the implemented retrieval pipeline.

Algorithm Working

From a question to a ranked Top-10 list

A Normalize query

B Detect intent

C Retrieve candidates

D Normalize scores

E Fuse signals

F Rerank Top 50

G Filter versions / duplicates

H Return Top 10

HYBRID SCORE

$$\text{Final} = w_1 \cdot \text{BM25} + w_2 \cdot \text{Semantic} + w_3 \cdot \text{Structure} + w_4 \cdot \text{Intent}$$

PSEUDOCODE

```
candidates ← BM25(query) ∪ Dense(query) ∪ Structure(query)
Top50 ← normalize_and_fuse(candidates, intent)
reranked ← rerank(query, Top50)
final ← diversify_and_filter(reranked)
return first 10(final)
```

Key idea: expensive ranking happens only after recall has reduced the search space.

Impact & Use Cases

A retrieval layer for developers and AI coding agents

DEVELOPER

Find implementation quickly

AI CODING AGENT

Ground reasoning in real code

MAINTAINER

Trace versions and behaviour

ONE SEARCH → MANY DEVELOPMENT TASKS

Validation

Find checks before persistence

Auth

Locate token/session validation

Preprocessing

Trace normalize / trim / parse

Errors

Find exception handling

API lookup

Find behaviour, not just names

RIGHT CODE → BETTER CONTEXT → MORE RELIABLE DOWNSTREAM AI

Innovation, Results & Limitations

What is differentiated — and what must still be measured

01
INTENT-AWARE

Query intent changes retrieval weighting

02
STRUCTURE-AWARE

Calls, operations, parameters and returns

03
TWO-STAGE RANKING

Top-50 reranking controls compute cost

04
EVOLUTION-AWARE

Versions, duplicates and lineages

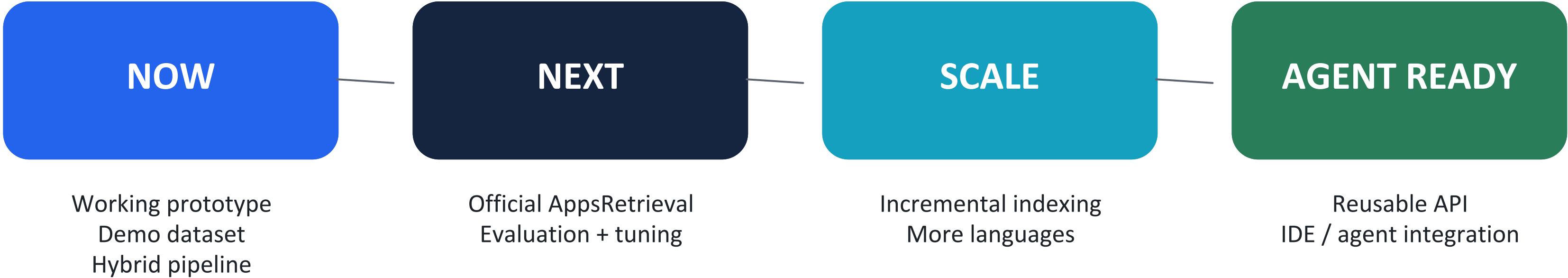
Experiment	What it isolates	Metric
BM25	Lexical baseline	NDCG@10 / MRR
Dense	Semantic baseline	NDCG@10 / MRR
Hybrid	Lexical + semantic	NDCG@10 / MRR
Hybrid + intent	Query-aware weighting	NDCG@10 / MRR
Final pipeline	Structure + rerank + filtering	NDCG@10 / MRR

LIMITATIONS

- Benchmark results require the official evaluation set.
- Heavy rerankers may need CPU optimization.
- Version lineage depends on metadata.
- Dataset quality affects retrieval quality.

What's Next

Roadmap from working prototype to robust repository-scale retrieval



ENGINEERING PRIORITIES

- ✓ Tune weights against NDCG@10 / MRR
- ✓ Strengthen version lineage
- ✓ Expand difficult cross-file queries
- ✓ Improve structural parsing
- ✓ Optimize latency + caching

Brownie Points: Differentiation

Four complementary views of the same codebase



Retrieve wide → combine evidence → rerank narrow → diversify

Submission Checklist

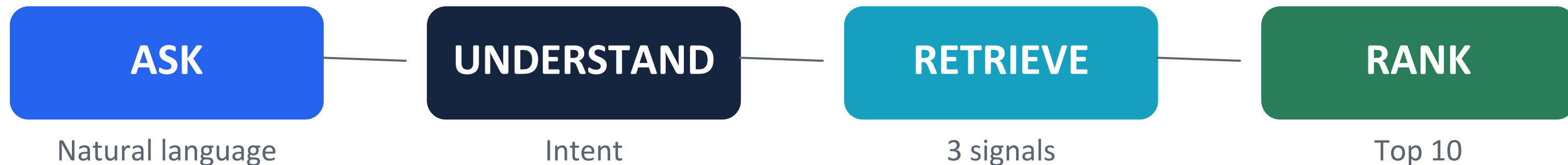
Make the prototype reproducible and easy to evaluate

Deliverable	Status	What to verify
Working prototype	READY	End-to-end query → Top 10
README	READY	Setup, run, dataset, evaluation
Public GitHub	ADD LINK	Repository accessible to judges
Demo video ≤ 5 min	ADD LINK	Show real query + ranking
PPT / PDF	READY	Architecture + evidence + limitations
Evaluation artifact	IN PROGRESS	Official benchmark results when available



AGENTIC CODE INTELLIGENCE

Intent-aware hybrid retrieval for finding the right code before AI agents reason over it.



Thank you...